

Quantum harmonic oscillator

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- Eigenfunctions and Eigenvalues
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What you need to know*

- **Quantization of Vibrations in Molecules:** Vibrational degrees of freedom are quantized, with important implications for infrared and Raman spectroscopies, as well as molecular bonding.
- **Existence of Zero-Point Energy and Tunneling:** Energy is non-zero even at absolute zero temperature, which we will understand as a consequence of the uncertainty principle.
- **Hermite Polynomials as Eigenfunctions:** We will explore how Hermite polynomials serve as the eigenfunctions of the quantum harmonic oscillator.
- **Effects of Anharmonicity:** Moving beyond the harmonic approximation, we will examine how anharmonicity impacts energy levels in oscillators.
- **Raising and Lowering Operators:** These operators offer an elegant method for solving the harmonic oscillator problem, marking our first introduction to a powerful tool used throughout quantum mechanics.

Hamitlonian of Harmonic Oscillator

- Taking the classical Hamiltonian for a harmonic oscillator is given by:

$$H = \frac{1}{2\mu} p_x^2 + \frac{1}{2} kx^2$$

- The quantum mechanical harmonic oscillator is obtained by replacing the classical position and momentum by the corresponding quantum mechanical operators

$$\hat{H} = -\frac{\hbar^2}{2\mu} \frac{d^2}{dx^2} + \frac{1}{2} kx^2$$

- Now the problem is to solve for eignefunctions and eigenvalues.

$$\hat{H}\psi_v(x) = E_v\psi_v(x)$$

Parameters of the Hamiltonian

- $k = \frac{F}{x}$ Spring constant
- Angular frequency of vibrations $\omega = 2\pi\nu$
- ν , Frequency (Hz; do not confuse this with quantum number v)

Eigenfunctions and Eiganvalues

Eigenfunctions/Eiganvalues of HO

$$E_v = \left(v + \frac{1}{2} \right) \hbar\omega$$

$$\psi_v = N_v \cdot H_v \left(\sqrt{\alpha} x \right) \cdot e^{-\alpha x^2/2}$$

$$v = 0, 1, 2, 3 \dots$$

- **Scaling factor**, $\alpha = \sqrt{\frac{2\mu}{\hbar^2}}$
- **Normalization factor**, $N_v = \frac{1}{\sqrt{2^v v!}} \left(\frac{\alpha}{\pi} \right)^{1/4}$
- **Hermite polynomials**, H_v

- For example, the wavefunctions for the two lowest states are:

$$\psi_0(x) = \left(\frac{\alpha}{\pi} \right)^{1/4} e^{-\alpha x^2/2}$$

$$\psi_1(x) = \left(\frac{4\alpha^3}{\pi} \right)^{1/4} x e^{-\alpha x^2/2}$$

Hermite polynomials show nodal and symmetry features of wavefunctions

n	Hermite Polynomial ($H_n(x)$)
0	1
1	$2x$
2	$4x^2 - 2$
3	$8x^3 - 12x$
4	$16x^4 - 48x^2 + 12$

Hermite polynomials obey the following relations which are useful when evaluating integrals.

Characteristic Equation

$$H_v''(y) - 2yH_v'(y) + 2vH_v(y) = 0$$

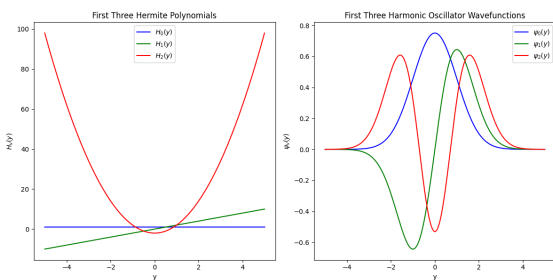
Recursion Relation

$$H_{v+1}(y) = 2yH_v(y) - 2vH_{v-1}(y)$$

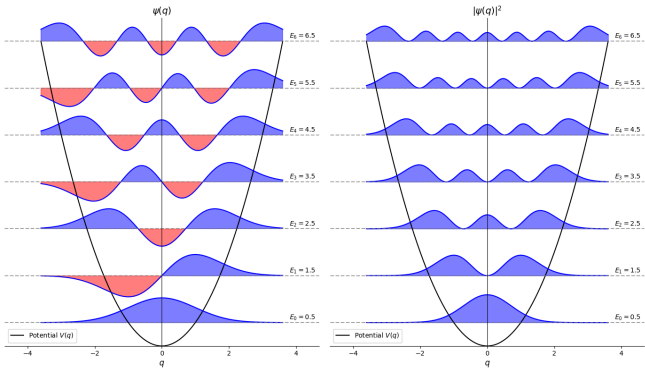
Orthogonality and Normalization

$$\int_{-\infty}^{\infty} H_v(y) H_{v'}(y) e^{-y^2} dy = \begin{cases} 0, & \text{if } v' \neq v \\ \sqrt{\pi} 2^v v!, & \text{if } v' = v \end{cases}$$

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Odd/even symmetry of wavefunctions

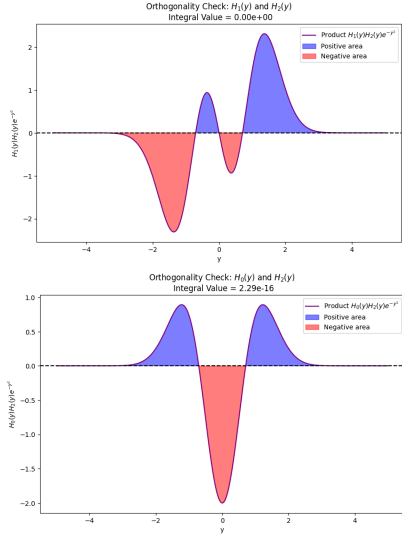
- Solutions ψ_v with $v = 0, 2, 4, \dots$ are even: $\psi_v(x) = \psi_v(-x)$.
- Solutions ψ_v with $v = 1, 3, 5, \dots$ are odd: $\psi_v(x) = -\psi_v(-x)$.

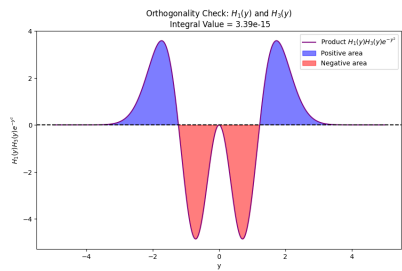
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Consequences for evaluating integrals

- Integral of an odd function from $-a$ to a (a may be ∞) is zero.

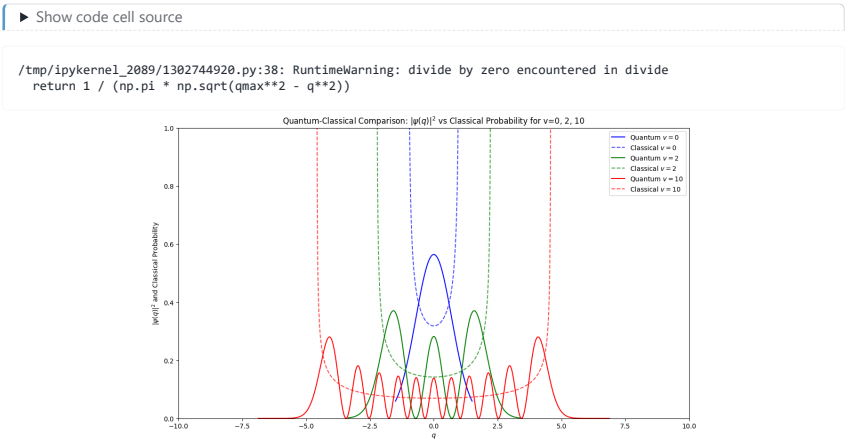
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Interpreting solutions

- The tails of the wavefunctions penetrate into the potential barrier deeper than the classical physics would allow. This phenomenon is called **tunneling**.
- As we excite the system to higher levels probability distribution accumulates near edges thereby approaching classical prediction.



Problems

Problem 1

Show that the lowest level of Harmonic oscillator obeys the uncertainty principle.

Solution

$$\Delta x = \sigma_x = \sqrt{\langle \hat{x}^2 \rangle - \langle \hat{x} \rangle^2} \text{ and } \Delta p_x = \sigma_{p_x} = \sqrt{\langle \hat{p}_x^2 \rangle - \langle \hat{p}_x \rangle^2}$$

First we calculate $\langle \hat{x} \rangle$ (ψ_0 is an even function, x is odd, the integrand is odd overall):

$$\langle \hat{x} \rangle = \int_{-\infty}^{\infty} \psi_0(x) x \psi_0(x) dx = 0$$

For $\langle \hat{x}^2 \rangle$ we have (integration by parts or tablebook):

$$\begin{aligned} \langle \hat{x}^2 \rangle &= \int_{-\infty}^{\infty} \psi_0(x) x^2 \psi_0(x) dx = \left(\frac{\alpha}{\pi} \right)^{1/2} \int_{-\infty}^{\infty} x^2 e^{-\alpha x^2} dx = \left(\frac{\alpha}{\pi} \right)^{1/2} \left[\frac{1}{2\alpha} \left(\frac{\pi}{\alpha} \right)^{1/2} \right] \\ &= \frac{1}{2\alpha} = \frac{1}{2} \frac{\hbar}{\sqrt{\mu k}} \Rightarrow \Delta x = \sqrt{\frac{1}{2} \frac{\hbar}{\sqrt{\mu k}}} \end{aligned}$$

For $\langle \hat{p}_x \rangle$ we have again by symmetry:

$$\langle \hat{p}_x \rangle = \int_{-\infty}^{\infty} \underbrace{\psi_0(x)}_{\text{even}} \underbrace{\left(-i\hbar \frac{d}{dx} \right) \psi_0(x)}_{\text{odd}} dx = 0$$

Note that derivative of an even function is an odd function. For $\langle \hat{p}_x^2 \rangle$ we have:

$$\begin{aligned} \langle \hat{p}_x^2 \rangle &= \int_{-\infty}^{\infty} \psi_0(x) \hat{p}_x^2 \psi_0(x) dx = -\hbar^2 \left(\frac{\alpha}{\pi} \right)^{1/2} \int_{-\infty}^{\infty} e^{-\alpha x^2/2} \frac{d^2}{dx^2} e^{-\alpha x^2/2} dx \\ &= \hbar^2 \left(\frac{\alpha}{\pi} \right)^{1/2} \int_{-\infty}^{\infty} (\alpha - \alpha^2 x^2) e^{-\alpha x^2} dx = \left[\hbar^2 \left(\frac{\alpha}{\pi} \right)^{1/2} \right] \\ &\quad \times \left(\alpha \int_{-\infty}^{\infty} e^{-\alpha x^2} dx - \alpha^2 \int_{-\infty}^{\infty} x^2 e^{-\alpha x^2} dx \right) \\ &= \left[\hbar^2 \left(\frac{\alpha}{\pi} \right)^{1/2} \right] \times \left(\alpha \sqrt{\frac{\pi}{\alpha}} - \alpha^2 \frac{\sqrt{\pi}}{2\alpha^{3/2}} \right) \\ &= \left[\hbar^2 \sqrt{\frac{\alpha}{\pi}} \right] \times \left(\sqrt{\pi\alpha} - \frac{\sqrt{\pi\alpha}}{2} \right) = \frac{\hbar^2 \alpha}{2} = \frac{\hbar \sqrt{\mu k}}{2} \Rightarrow \Delta p_x = \sqrt{\frac{\hbar \sqrt{\mu k}}{2}} \end{aligned}$$

Finally, we can calculate $\Delta x \Delta p_x$:

$$\Delta x \Delta p_x = \sqrt{\frac{1}{2} \frac{\hbar}{\sqrt{\mu k}}} \times \sqrt{\frac{\hbar \sqrt{\mu k}}{2}} = \sqrt{\frac{\hbar^2}{4}} = \frac{\hbar}{2}$$

Recall that the uncertainty principle stated that: $\Delta x \Delta p_x \geq \frac{\hbar}{2}$

Thus we can conclude that ψ_0 fulfils the Heisenberg uncertainty principle.

Problem 2

Quantization of nuclear motion. [Molecular vibration](#) in a diatomic molecule can be approximated by the quantum mechanical harmonic oscillator model. There μ is the reduced mass as given previously and the variable x is the distance between the atoms in the molecule (or more exactly, the deviation from the equilibrium bond length R_e).

- a. Derive the expression for the standard deviation of the bond length in a diatomic molecule when it is in its ground vibrational state.
- b. What percentage of the equilibrium bond length is this standard deviation for carbon monoxide in its ground vibrational state? For $^{12}C^{16}O$, we have: $\tilde{\nu} = 2170 \text{ cm}^{-1}$ (vibrational frequency) and $R_e = 113 \text{ pm}$ (equilibrium bond length)

Solution

The harmonic vibration frequency is given in wavenumber units (cm^{-1}). This must be converted according to: $\nu = c\tilde{\nu}$. The previous example gives expression for σ_x :

$$\sigma_x = \Delta x = \sqrt{\frac{1}{2} \frac{\hbar}{\sqrt{\mu k}}}$$

In considering spectroscopic data, it is convenient to express this in terms of $\tilde{\nu}$:

$$k = (2\pi c\tilde{\nu})^2 \mu \text{ and } \Delta x = \sigma_x = \sqrt{\frac{\hbar}{4\pi c\tilde{\nu} \mu}}$$

In part (b) we have to apply the above expression to find out the standard deviation for carbon monoxide bond length in its ground vibrational state. First we need the reduced mass:

$$\begin{aligned} \mu &= \frac{m_1 m_2}{m_1 + m_2} = \frac{(12 \times 10^{-3} \text{ kg mol}^{-1})(15.995 \times 10^{-3} \text{ kg mol}^{-1})}{((12 + 15.995) \times 10^{-3} \text{ kg mol}^{-1}) \underbrace{(6.022 \times 10^{23} \text{ mol}^{-1})}_{\text{Avogadro's constant}}} \\ &= 1.139 \times 10^{-26} \text{ kg} \end{aligned}$$

The standard deviation is now:

$$\begin{aligned} \Delta x = \sigma_x &= \left[\frac{1.055 \times 10^{-34} \text{ Js}}{4\pi \underbrace{(2.998 \times 10^{10} \text{ cm s}^{-1})}_{\text{speed of light}} (2170 \text{ cm}^{-1}) (1.139 \times 10^{-26} \text{ kg})} \right]^{1/2} \\ &= 3.37 \text{ pm} \Rightarrow \% \text{ of deviation} = 100\% \times \frac{3.37 \text{ pm}}{113 \text{ pm}} = 2.98\% \end{aligned}$$

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